Habib University
School of Science & Engineering

Course:	Linear Integrated Circuits
Semester	Fall 2025
Quiz #	2
Date, Time :	November 25, 2025, 1 hour
Instructor :	Syed Arsalan Jawed
Total Marks :	100

Syed Muhammad Asad Ali (sa09475)

Instructions for Students

1. The following design task needs to be completed within this lecture session and the required report uploaded to the LMS.
2. Either attach a clear drawn schematic from the tool or draw by hand, unclear and untidy schematics and unclear waveforms are not acceptable.
3. Zero tolerance to consultation and copying from the other students.

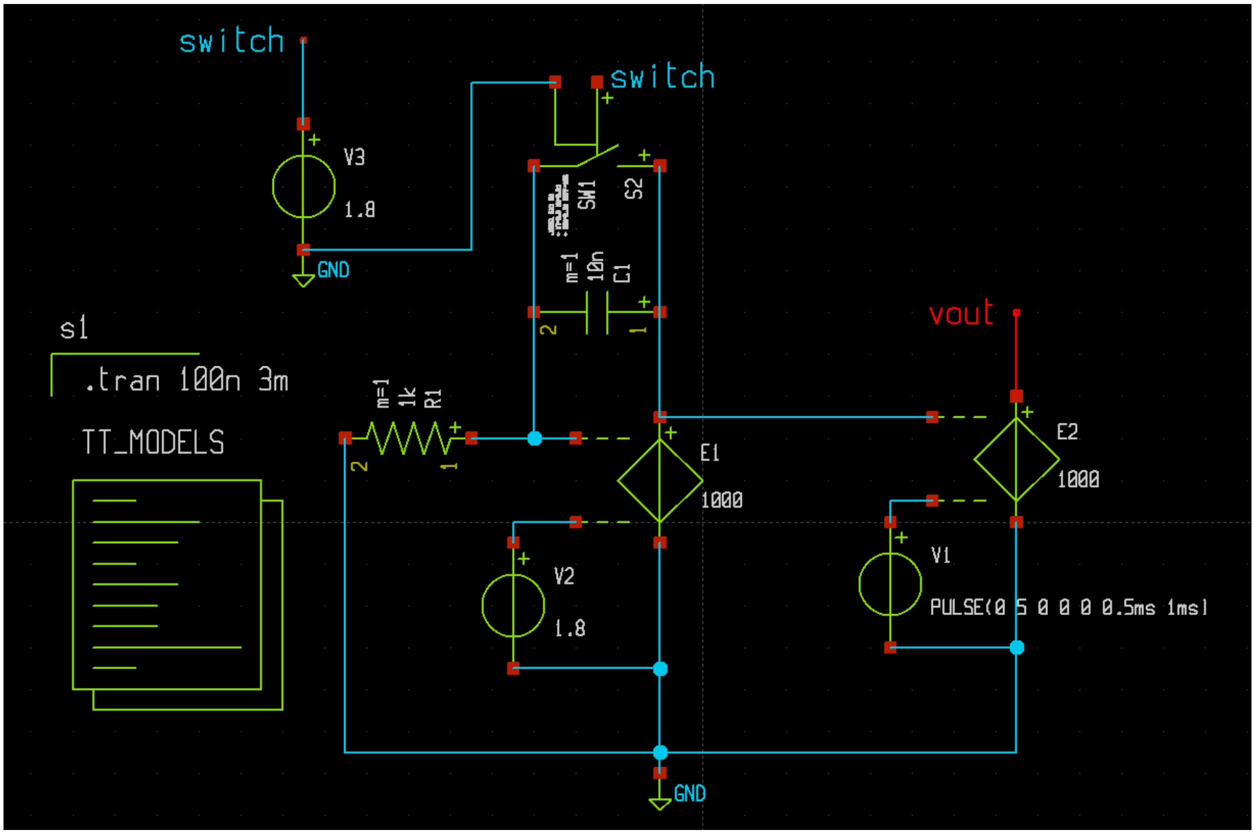
Question # 1 : [CLO-1,2,3] [50]

A single slope integrating ADC needs to be designed and simulated using Opamp based circuits. The counter part of the ADC does not need to be implemented. The width of COUNTER_ENABLE can be used as digital representation of the Analog Input.

1. The ADC should be able to convert signals till 1kHz, i.e., the analog bandwidth of the ADC is 1kHz.
2. Use supply voltage VDD of 1.8V. There is no negative supply. Rails are between VDD and GND(0V).
3. In place of Opamps, an ideal voltage-controlled-voltage-source (VCVS) is to be used.
4. The Analog input is three different dc levels : 0, VDD/2, VDD.
5. Take of the negative integrating slope by proper referencing of the VCVS.
6. Use a shorting switch along with a RESET signal generated through a voltage pulse source to reset the integrating capacitor at the start. You may use an ideal switch.

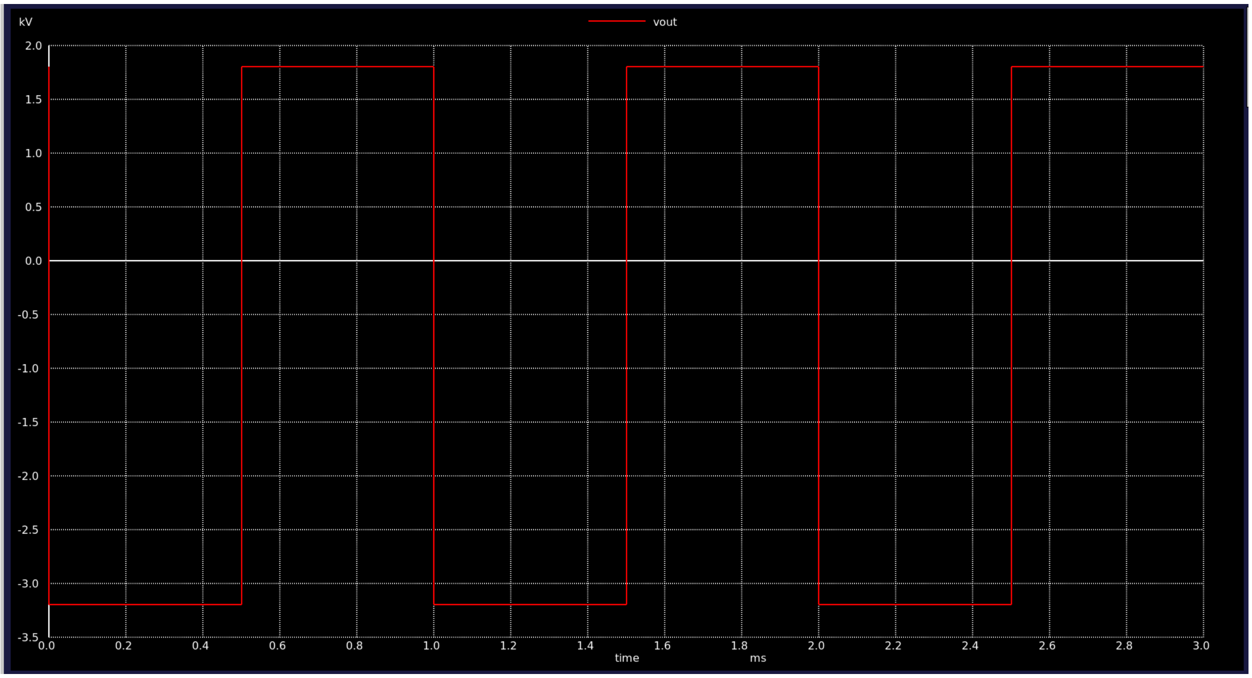
1) Simulate the circuit in XSCHEM/NGSPICE and fill page 2, 3 and 4 of this document.

Schematic Snap

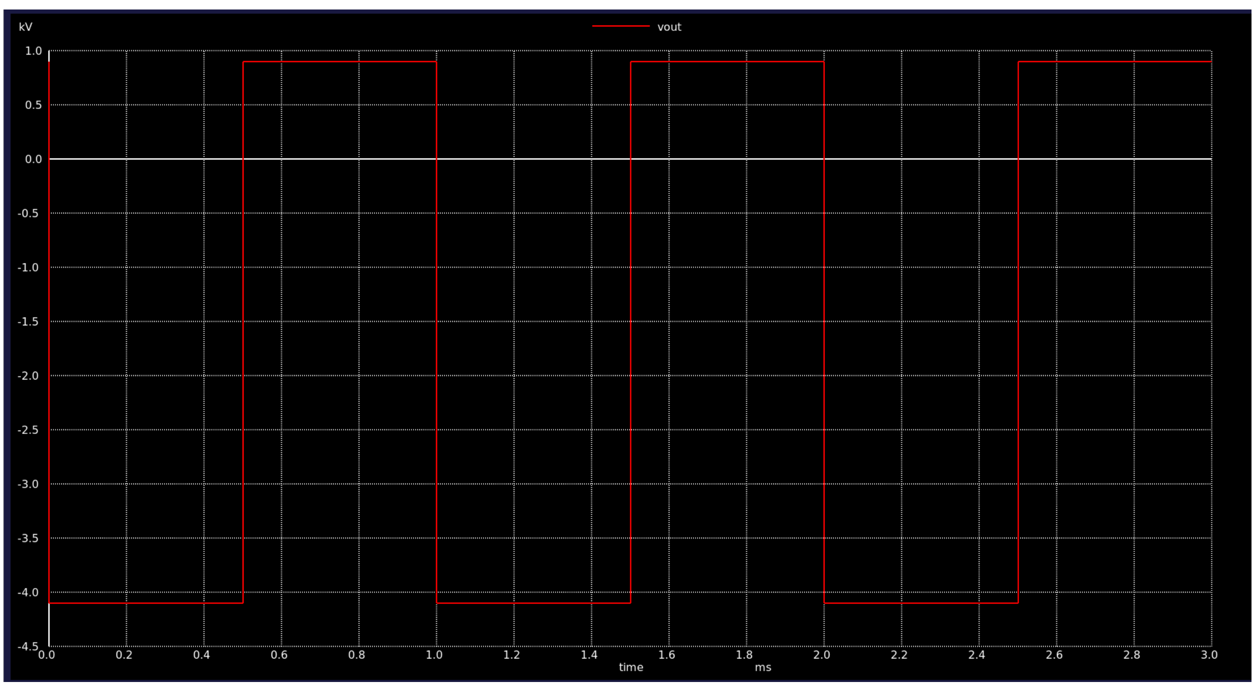


Waveforms

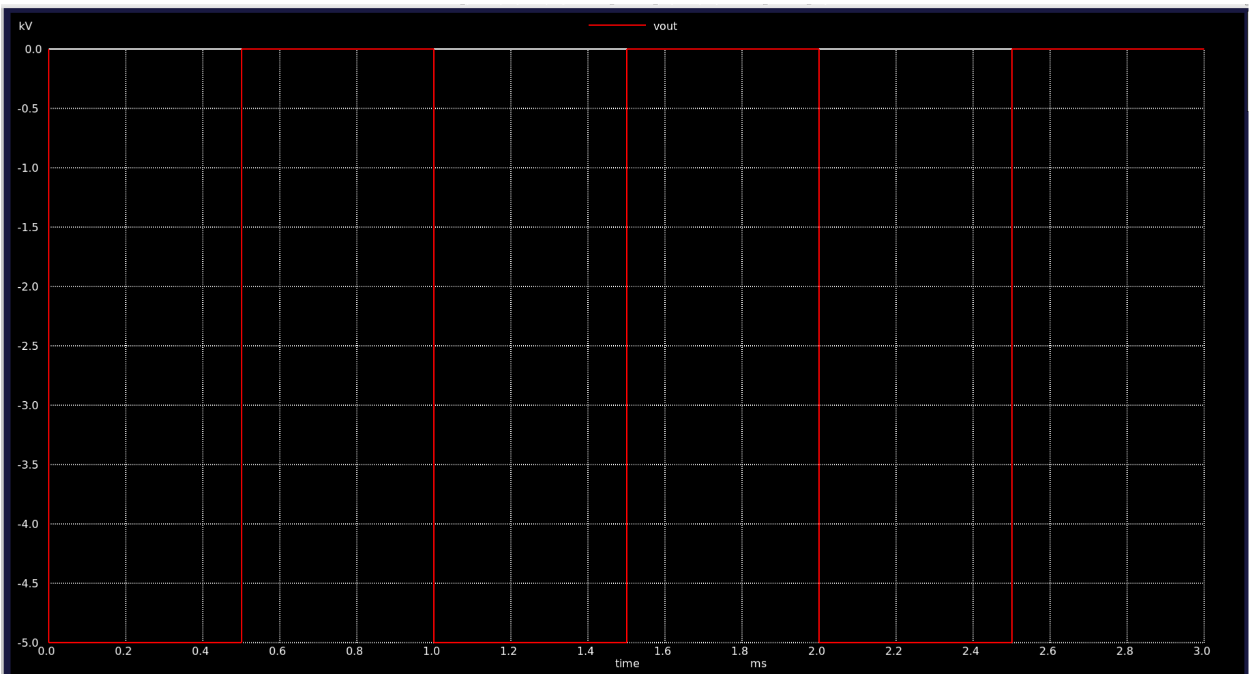
At Vref=1.8V



At Vref = 0.9V:



At V=0V:


kV
vout
0.0
-0.5
-1.0
-1.5
-2.0
-2.5
-3.0
-3.5
-4.0
-4.5
-5.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
2.4
2.6
2.8
3.0
time
ms

Supporting Explanation